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(54) **DYNAMIC INITIALIZATION OF 3DOF AR TRACKING SYSTEM**

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(57)

ABSTRACT

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G06F 3/0346 (2013.01)
G06F 3/038 (2013.01)
(52) **U.S. Cl.**
CPC **G06F 3/0346** (2013.01); **G06F 3/038** (2013.01)

A method for dynamically initializing a 3 degrees of freedom (3DOF) tracking device is described. In one aspect, the method includes accessing a gyroscope signal from a gyroscope of the 3DOF tracking device, accessing an accelerometer signal from an accelerometer of the 3DOF tracking device, determining an initial state includes a combination of an initial orientation, an initial position, and an initial velocity of the 3DOF tracking device, the initial state indicating a starting condition of the 3DOF tracking device, integrating the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state, and refining an inclination signal of the orientation signal using the position signal.

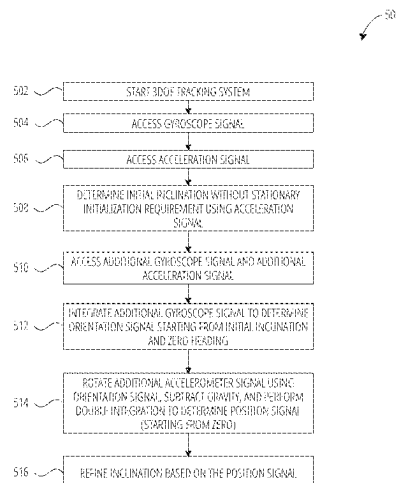
(58) **Field of Classification Search**
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See application file for complete search history.

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20 Claims, 10 Drawing Sheets



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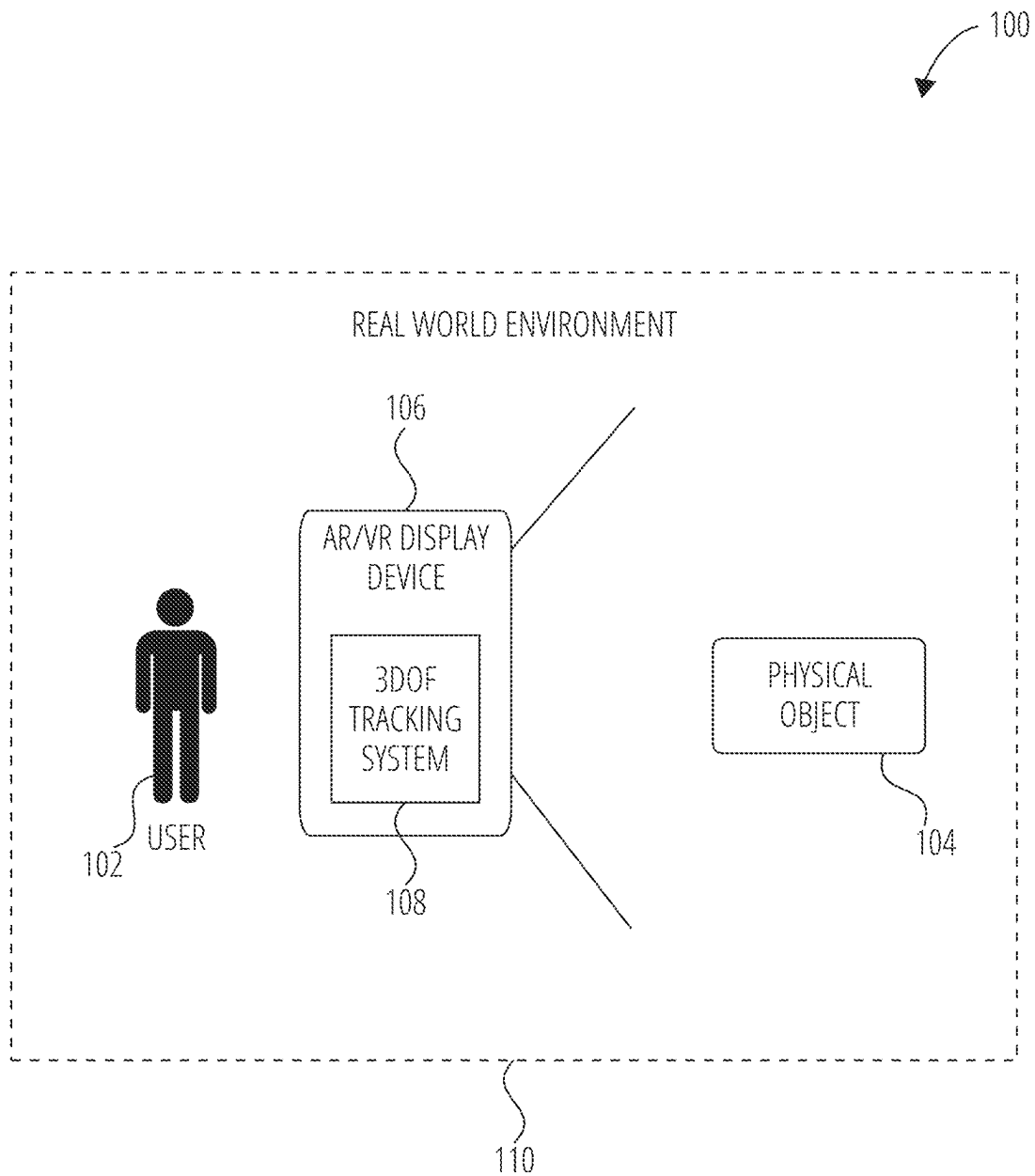


FIG. 1

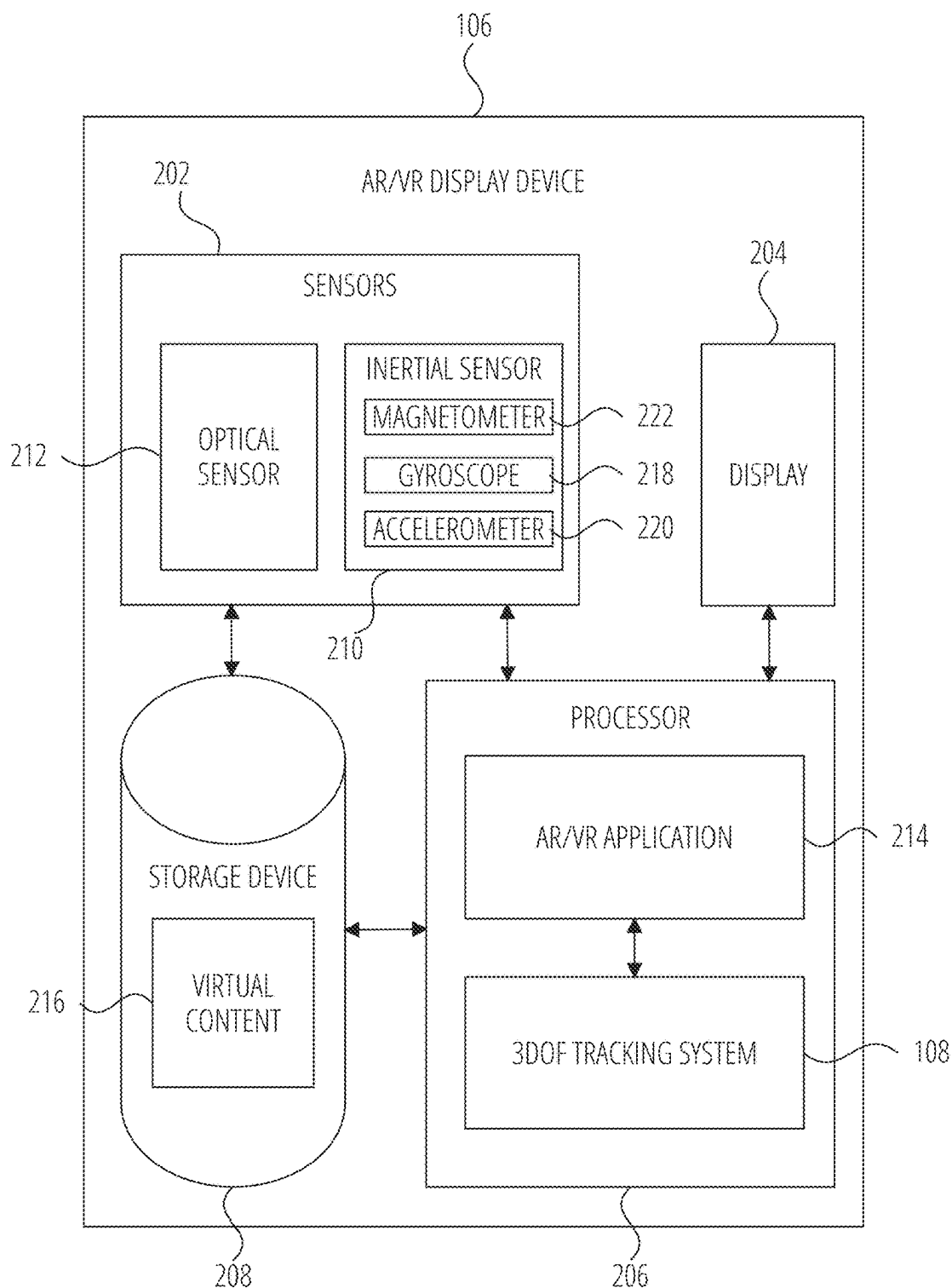


FIG. 2

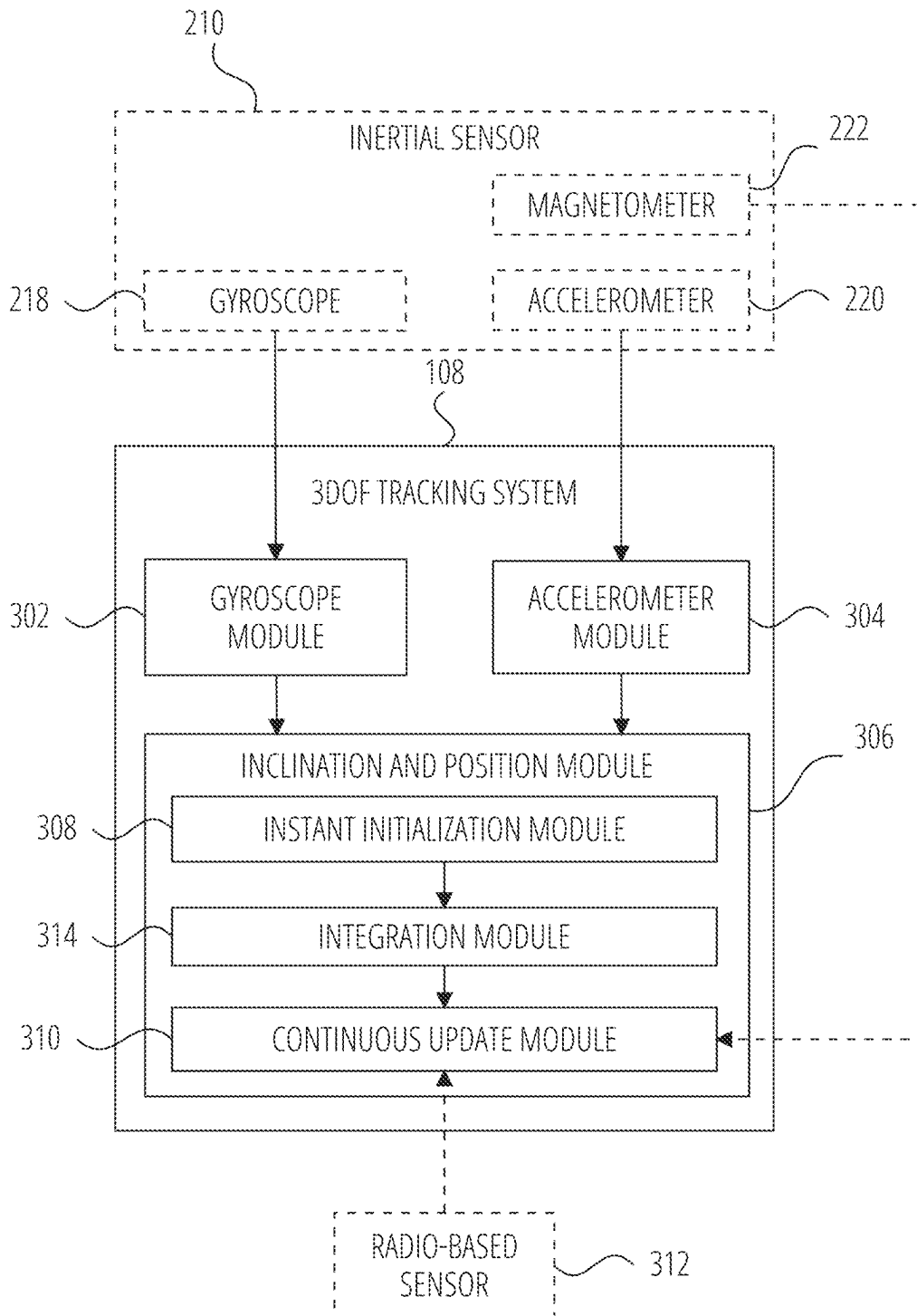


FIG. 3

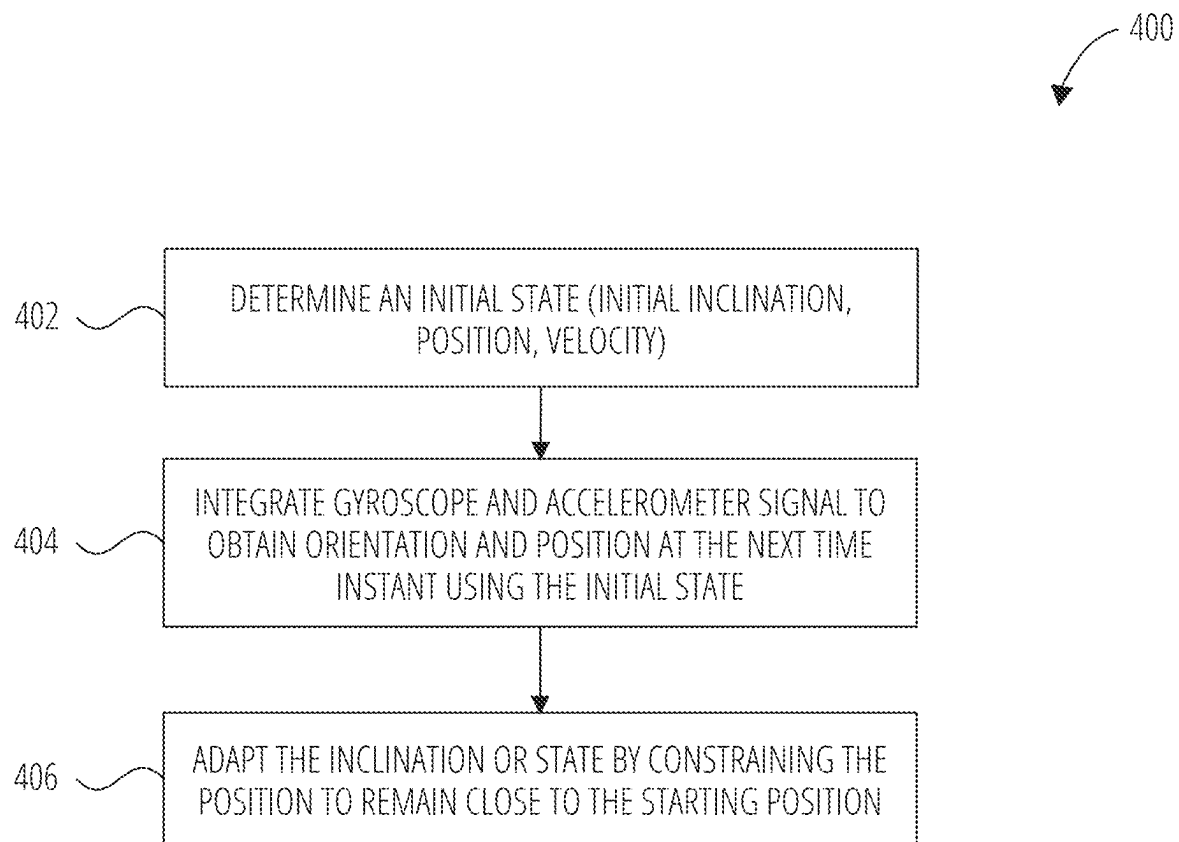
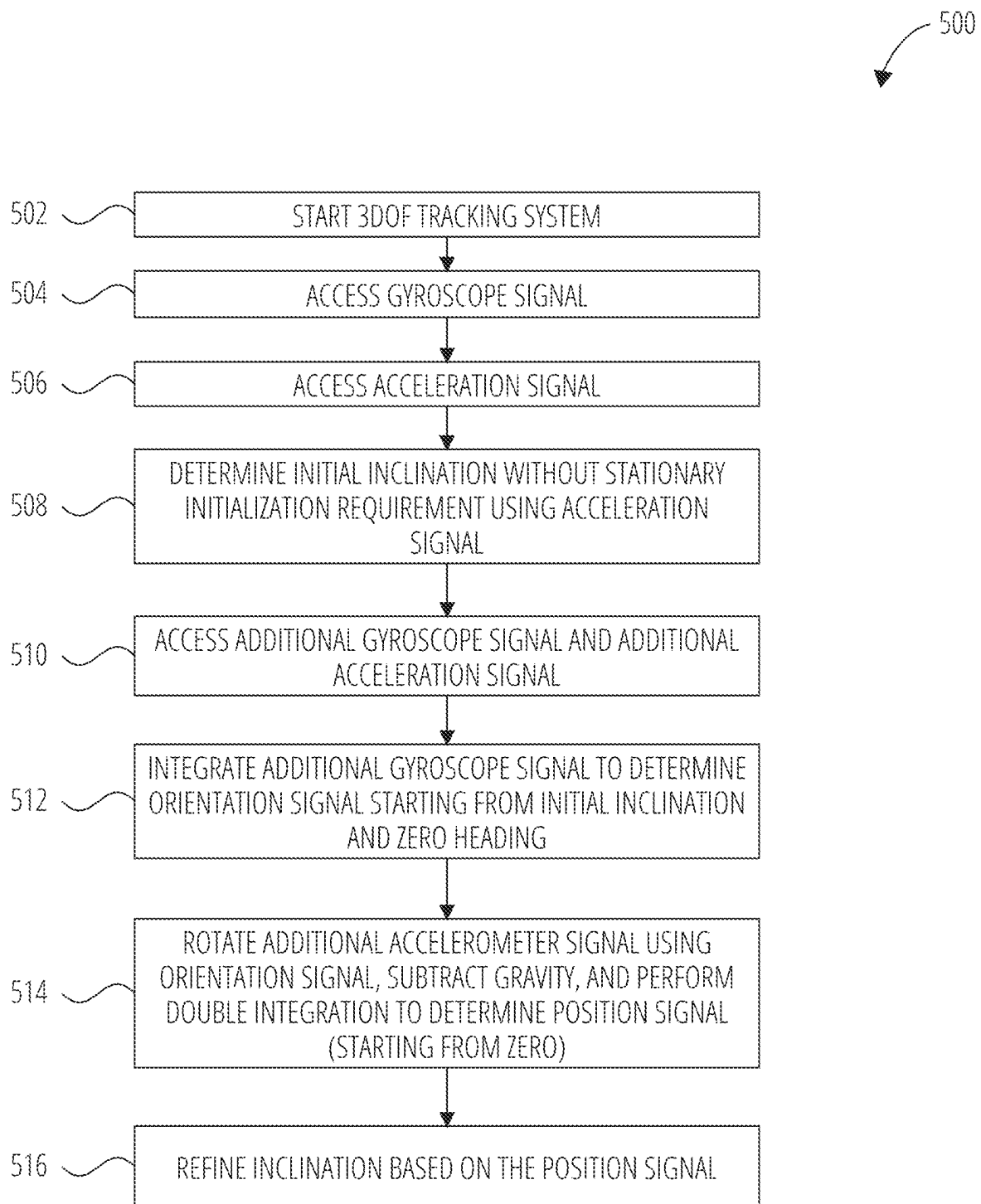


FIG. 4

**FIG. 5**

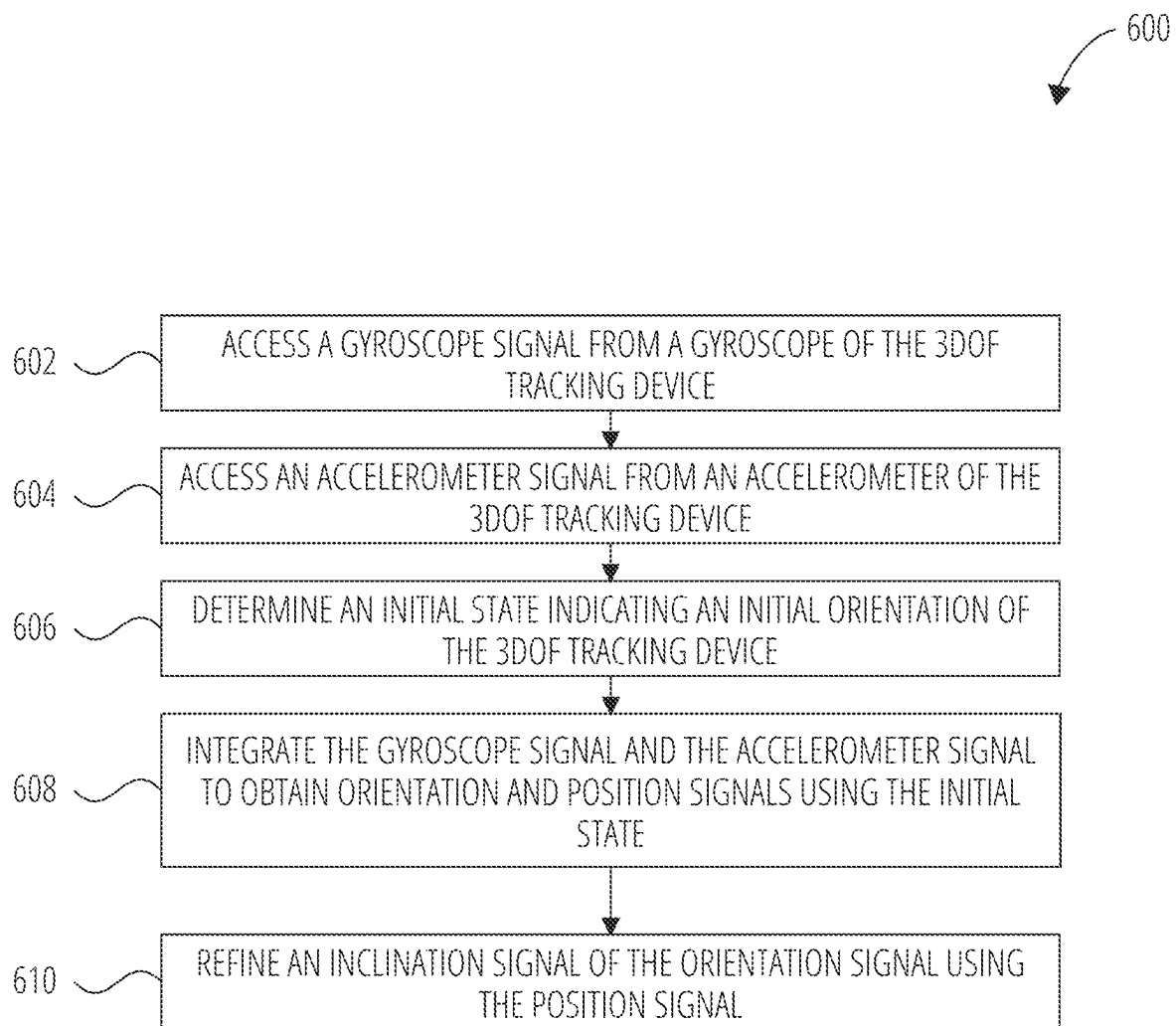


FIG. 6

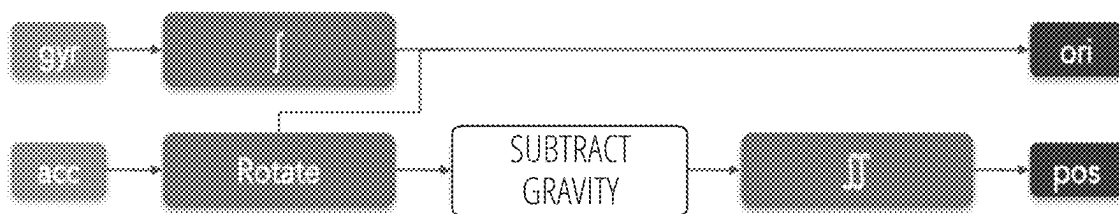


FIG. 7

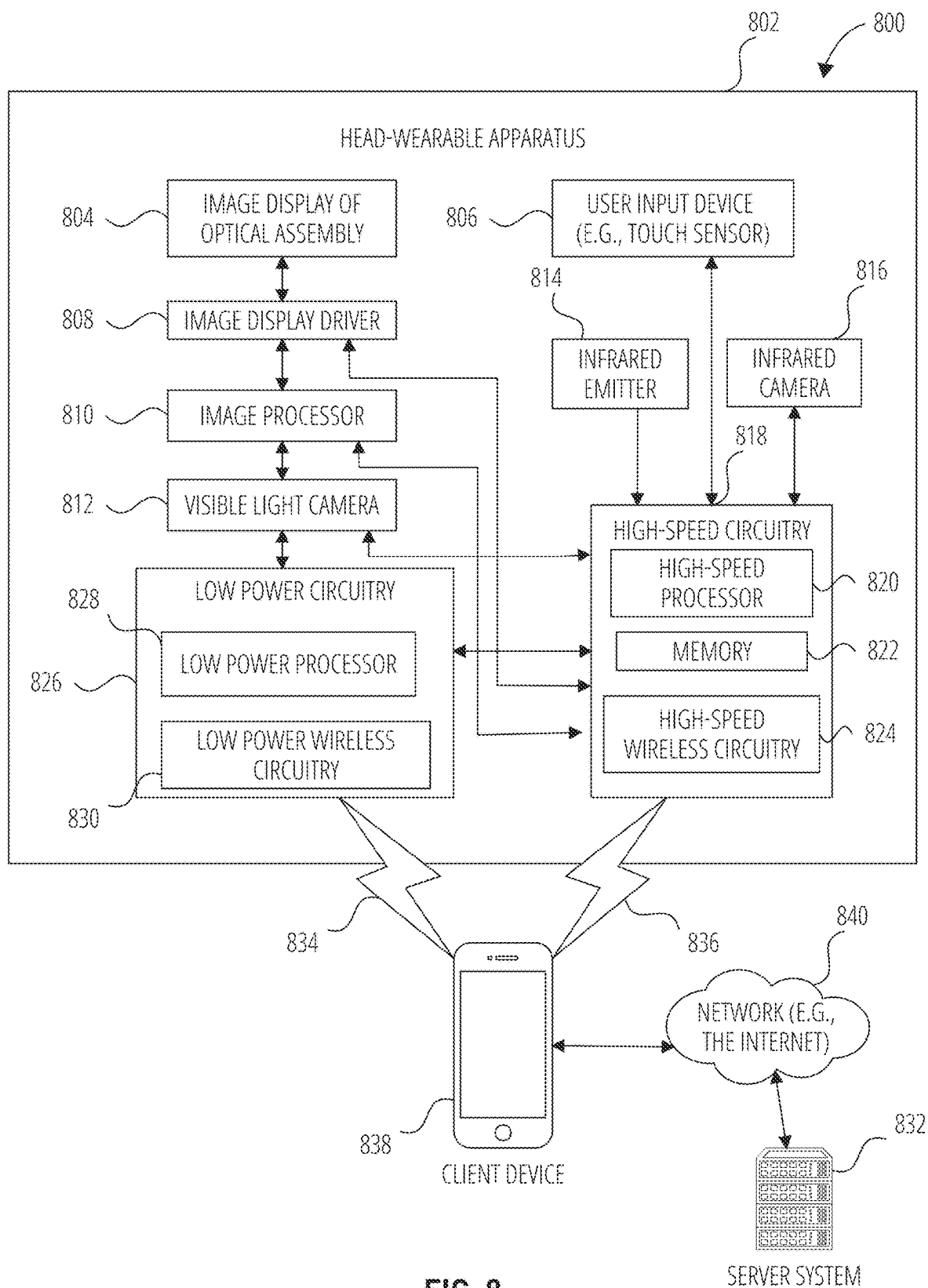
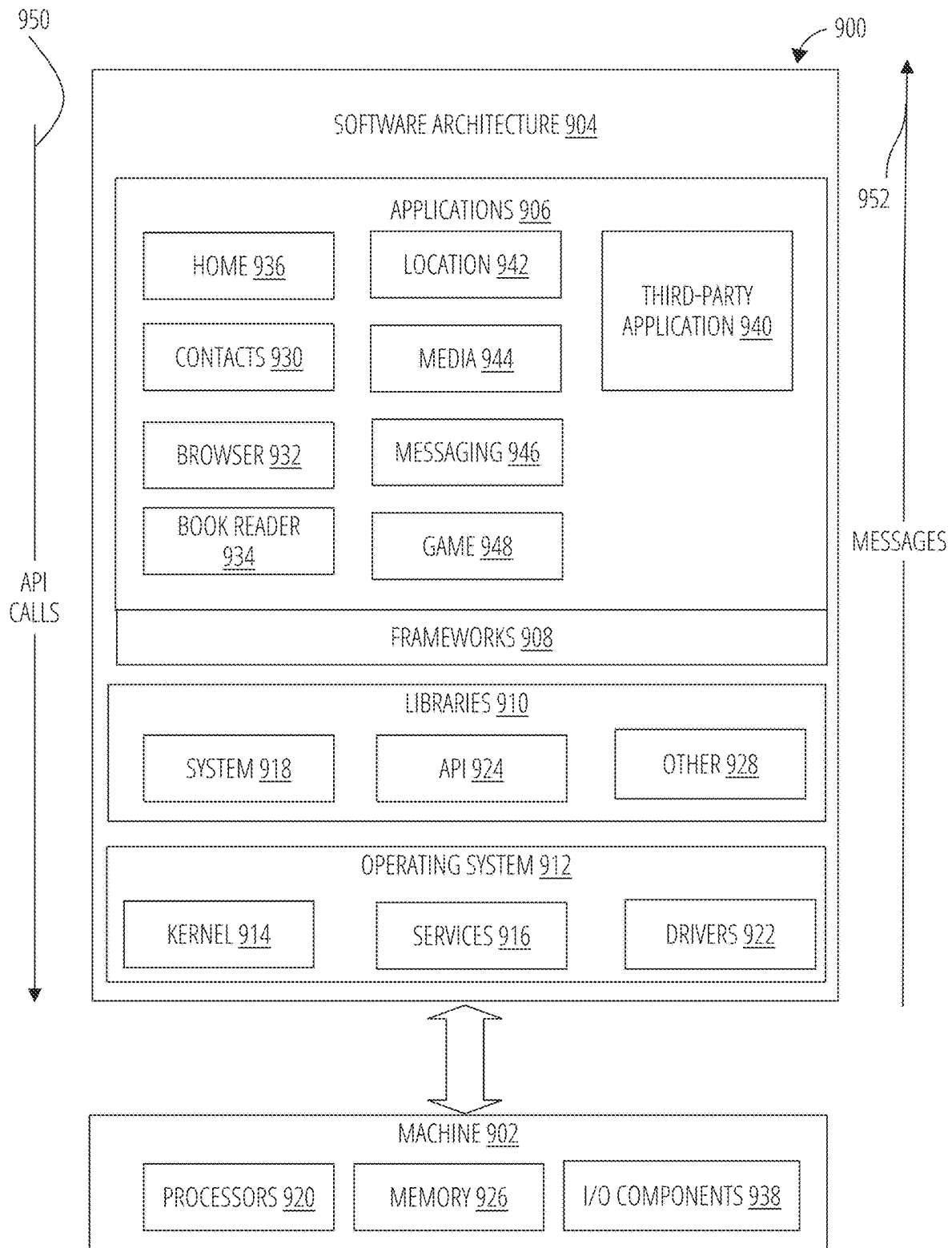


FIG. 8

**FIG. 9**

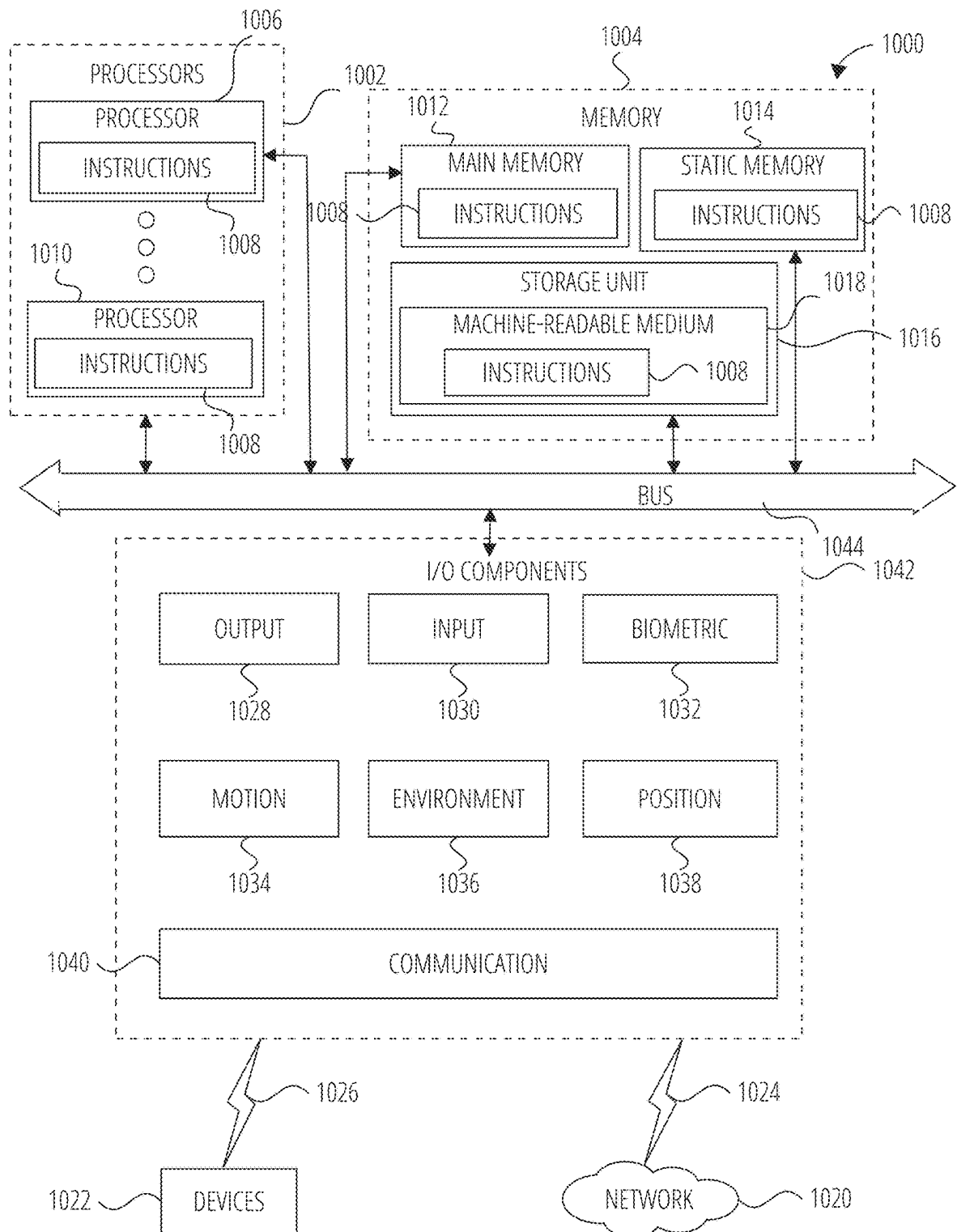


FIG. 10

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DYNAMIC INITIALIZATION OF 3DOF AR TRACKING SYSTEM

RELATED APPLICATION

The present application is a continuation of U.S. patent application Ser. No. 17/524,423, filed Nov. 11, 2021, which claims priority to U.S. Provisional Patent Application Ser. No. 63/189,845, filed May 18, 2021, which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

The subject matter disclosed herein generally relates to a visual tracking system. Specifically, the present disclosure addresses systems and methods for calibrating visual-inertial tracking systems.

BACKGROUND

An augmented reality (AR) device enables a user to observe a scene while simultaneously seeing relevant virtual content that may be aligned to items, images, objects, or environments in the field of view of the device. A virtual reality (VR) device provides a more immersive experience than an AR device. The VR device blocks out the field of view of the user with virtual content that is displayed based on a position and orientation of the VR device.

Both AR and VR devices rely on motion tracking system that track a pose (e.g., orientation, position, location) of the device. The motion tracking system is typically factory calibrated (based on predefined relative positions between the cameras and other sensors) to accurately display the virtual content at a desired location relative to its environment.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

FIG. 1 is a block diagram illustrating an environment for operating an AR/VR display device in accordance with one example embodiment.

FIG. 2 is a block diagram illustrating an AR/VR display device in accordance with one example embodiment.

FIG. 3 is a block diagram illustrating a 3DOF tracking system in accordance with one example embodiment.

FIG. 4 is a flow diagram illustrating a method for adapting the inclination in accordance with one example embodiment.

FIG. 5 is a flow diagram illustrating a method for refining an inclination signal in accordance with one example embodiment.

FIG. 6 illustrates a routine 600 in accordance with one embodiment.

FIG. 7 is a block diagram illustrating an example algorithm for initialization in accordance with one example embodiment.

FIG. 8 illustrates a network environment in which a head-wearable device can be implemented according to one example embodiment.

FIG. 9 is block diagram showing a software architecture within which the present disclosure may be implemented, according to an example embodiment.

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FIG. 10 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, according to one example embodiment.

DETAILED DESCRIPTION

The description that follows describes systems, methods, techniques, instruction sequences, and computing machine program products that illustrate example embodiments of the present subject matter. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the present subject matter. It will be evident, however, to those skilled in the art, that embodiments of the present subject matter may be practiced without some or other of these specific details. Examples merely typify possible variations. Unless explicitly stated otherwise, structures (e.g., structural components, such as modules) are optional and may be combined or subdivided, and operations (e.g., in a procedure, algorithm, or other function) may vary in sequence or be combined or subdivided.

The term “augmented reality” (AR) is used herein to refer to an interactive experience of a real-world environment where physical objects that reside in the real-world are “augmented” or enhanced by computer-generated digital content (also referred to as virtual content or synthetic content). AR can also refer to a system that enables a combination of real and virtual worlds, real-time interaction, and 3D registration of virtual and real objects. A user of an AR system perceives virtual content that appears to be attached or interact with a real-world physical object.

The term “virtual reality” (VR) is used herein to refer to a simulation experience of a virtual world environment that is completely distinct from the real-world environment. Computer-generated digital content is displayed in the virtual world environment. VR also refers to a system that enables a user of a VR system to be completely immersed in the virtual world environment and to interact with virtual objects presented in the virtual world environment.

The term “AR application” is used herein to refer to a computer-operated application that enables an AR experience. The term “VR application” is used herein to refer to a computer-operated application that enables a VR experience. The term “AR/VR application” refers to a computer-operated application that enables a combination of an AR experience or a VR experience.

The term “visual tracking system” is used herein to refer to a computer-operated application or system that enables a system to track visual features identified in images captured by one or more cameras of the visual tracking system. The visual tracking system builds a model of a real-world environment based on the tracked visual features. Non-limiting examples of the visual tracking system include: a visual Simultaneous Localization and Mapping system (VSLAM), and Visual Inertial Odometry (VIO) system. VSLAM can be used to build a target from an environment, or a scene based on one or more cameras of the visual tracking system. A VIO system (also referred to as a visual-inertial tracking system) determines a latest pose (e.g., position and orientation) of a device based on data acquired from multiple sensors (e.g., optical sensors, inertial sensors) of the device.

The term “Inertial Measurement Unit” (IMU) is used herein to refer to a device that can report on the inertial status of a moving body including the acceleration, velocity,

orientation, and position of the moving body. An IMU enables tracking of movement of a body by integrating the acceleration and the angular velocity measured by the IMU. IMU can also refer to a combination of accelerometers and gyroscopes that can determine and quantify linear acceleration and angular velocity, respectively. The values obtained from the IMUs gyroscopes can be processed to obtain the pitch, roll, and heading of the IMU and, therefore, of the body with which the IMU is associated. Signals from the IMU's accelerometers also can be processed to obtain velocity and displacement of the IMU.

The term “three-degrees of freedom tracking system” (3DOF tracking system) is used herein to refer to a device that tracks rotational movement. For example, the 3DOF tracking system can track whether a user of a head-wearable device is looking left or right, rotating their head up or down, and pivoting left or right. However, the head-wearable device cannot use the 3DOF tracking system to determine whether the user has moved around a scene by moving in the physical world. As such, 3DOF tracking system may not be accurate enough to be used for positional signals. The 3DOF tracking system may be part of an AR/VR display device that includes IMU sensors. For example, the 3DOF tracking system uses sensor data from sensors such as accelerometers, gyroscopes, and magnetometers.

The term “six-degrees of freedom tracking system” (6DOF tracking system) is used herein to refer to a device that tracks rotational and translational motion. For example, the 6DOF tracking system can track whether the user has rotated their head and moved forward or backward, laterally or vertically and up or down. The 6DOF tracking system may include a visual odometry system that relies on data acquired from multiple sensors (e.g., depth cameras, inertial sensors). The 6DOF tracking system analyzes data from the sensors to accurately determine the pose of the display device.

Both AR and VR applications allow a user to access information, such as in the form of virtual content rendered in a display of an AR/VR display device (also referred to as a display device). The rendering of the virtual content may be based on a position of the display device relative to a physical object or relative to a frame of reference (external to the display device) so that the virtual content correctly appears in the display. For AR, the virtual content appears aligned with a physical object as perceived by the user and a camera of the AR display device. The virtual content appears to be attached to the physical world (e.g., a physical object of interest). To do this, the AR display device detects the physical object and tracks a pose of the AR display device relative to the position of the physical object. A pose identifies a position and orientation of the display device relative to a frame of reference or relative to another object. For VR, the virtual object appears at a location based on the pose of the VR display device. The virtual content is therefore refreshed based on the latest pose of the device. A visual tracking system at the display device determines the pose of the display device. An example of a visual tracking system includes a visual-inertial tracking system (also referred to as VIO system) that relies on data acquired from multiple sensors (e.g., optical sensors, inertial sensors).

Initialization of a 3DOF tracking system typically requires the tracking system to be stationary in order to calibrate the sensors in the 3DOF tracking system. For example, the 3DOF tracking system detects that an AR/VR display device is turned/switched on. In response, the AR/VR display device requests the user to set the AR/VR display device on a stationary support (e.g., a table, a chair).

Such initialization process creates friction for the user to operate the AR/VR display device efficiently. Furthermore, such initialization process requires additional time to initialize under the stationary conditions.

The present application describes a method for dynamic initialization of a 3DOF tracking system of an AR/VR display device that allows the AR/VR display device to instantly initialize and starts a 3DOF (gravity aligned) tracking experience under dynamic and non-stationary conditions. In one example, the present method relies on the assumption that people stay roughly on the same spot is valid for most use cases (sitting, walking). As such, this removes the requirement that the user needs to remain still.

In one example embodiment, the method includes collecting accelerometer and gyroscope signals, integrating to orientation and position signals, constraining the position signal to remain ‘close’ to the (zero) starting position (which enable displaying of gravity aligned content in a head-worn AR system and do not require the user to be stationary while initializing).

In another example embodiment, a method for dynamically initializing a 3 degrees of freedom (3DOF) tracking device is described. In one aspect, the method includes accessing a gyroscope signal from a gyroscope of the 3DOF tracking device, accessing an accelerometer signal from an accelerometer of the 3DOF tracking device, determining an initial state includes a combination of an initial orientation, an initial position, and an initial velocity of the 3DOF tracking device, the initial state indicating a starting condition of the 3DOF tracking device, integrating the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state, and refining an inclination signal of the orientation signal using the position signal.

As a result, one or more of the methodologies described herein facilitate solving the technical problem of efficient initialization without requiring stationary conditions. The presently described method provides an improvement to an operation of the functioning of a computer by faster calibration computation. As such, one or more of the methodologies described herein may obviate a need for certain efforts or computing resources. Examples of such computing resources include Processor cycles, network traffic, memory usage, data storage capacity, power consumption, network bandwidth, and cooling capacity.

FIG. 1 is a network diagram illustrating an environment 100 suitable for operating an AR/VR display device 106, according to some example embodiments. The environment 100 includes a user 102, an AR/VR display device 106, and a physical object 104. A user 102 operates the AR/VR display device 106. The user 102 may be a human user (e.g., a human being), a machine user (e.g., a computer configured by a software program to interact with the AR/VR display device 106), or any suitable combination thereof (e.g., a human assisted by a machine or a machine supervised by a human). The user 102 is associated with the AR/VR display device 106.

The AR/VR display device 106 may be a computing device with a display such as a smartphone, a tablet computer, or a wearable computing device (e.g., watch or glasses). The computing device may be hand-held or may be removable mounted to a head of the user 102. In one example, the display includes a screen that displays images captured with a camera of the AR/VR display device 106. In another example, the display of the device may be transparent such as in lenses of wearable computing glasses. In other examples, the display may be non-transparent, partially

transparent, partially opaque. In yet other examples, the display may be wearable by the user **102** to cover the field of vision of the user **102**.

The AR/VR display device **106** includes an AR application that generates virtual content based on images detected with the camera of the AR/VR display device **106**. For example, the user **102** may point a camera of the AR/VR display device **106** to capture an image of the physical object **104**. The AR application generates virtual content corresponding to an identified object (e.g., physical object **104**) in the image and presents the virtual content in a display of the AR/VR display device **106**.

The AR/VR display device **106** includes a 3DOF tracking system **108**. The 3DOF tracking system **108** tracks a rotational pose (e.g., orientation) of the AR/VR display device **106** relative to the real world environment **110** using, for example, optical sensors (e.g., depth-enabled 3D camera, image camera), inertial sensors (e.g., gyroscope, accelerometer), wireless sensors (Bluetooth, Wi-Fi), GPS sensor, and audio sensor. In one example, the AR/VR display device **106** displays virtual content based on the pose of the AR/VR display device **106** relative to the real world environment **110** and/or the physical object **104**.

Any of the machines, databases, or devices shown in FIG. **1** may be implemented in a general-purpose computer modified (e.g., configured or programmed) by software to be a special-purpose computer to perform one or more of the functions described herein for that machine, database, or device. For example, a computer system able to implement any one or more of the methodologies described herein is discussed below with respect to FIG. **4** to FIG. **5**. As used herein, a “database” is a data storage resource and may store data structured as a text file, a table, a spreadsheet, a relational database (e.g., an object-relational database), a triple store, a hierarchical data store, or any suitable combination thereof. Moreover, any two or more of the machines, databases, or devices illustrated in FIG. **1** may be combined into a single machine, and the functions described herein for any single machine, database, or device may be subdivided among multiple machines, databases, or devices.

The AR/VR display device **106** may operate over a computer network. The computer network may be any network that enables communication between or among machines, databases, and devices. Accordingly, the computer network may be a wired network, a wireless network (e.g., a mobile or cellular network), or any suitable combination thereof. The computer network may include one or more portions that constitute a private network, a public network (e.g., the Internet), or any suitable combination thereof.

FIG. **2** is a block diagram illustrating modules (e.g., components) of the AR/VR display device **106**, according to some example embodiments. The AR/VR display device **106** includes sensors **202**, a display **204**, a processor **206**, and a storage device **208**. Examples of AR/VR display device **106** include a wearable computing device, a mobile computing device, a navigational device, a portable media device, or a smart phone.

The sensors **202** include, for example, an optical sensor **212** (e.g., camera such as a color camera, a thermal camera, a depth sensor and one or multiple grayscale, global/rolling shutter tracking cameras) and an inertial sensor **210** (e.g., gyroscope **218**, accelerometer **220**, magnetometer **222**). Other examples of sensors **202** include a proximity or location sensor (e.g., near-field communication, GPS, Bluetooth, Wifi, radio-based sensor **312**), an audio sensor (e.g., a microphone), a thermal sensor, a pressure sensor (e.g.,

barometer), or any suitable combination thereof. It is noted that the sensors **202** described herein are for illustration purposes and the sensors **202** are thus not limited to the ones described above.

The display **204** includes a screen or monitor configured to display images generated by the processor **206**. In one example embodiment, the display **204** may be transparent or semi-opaque so that the user **102** can see through the display **204** (in AR use case). In another example embodiment, the display **204** covers the eyes of the user **102** and blocks out the entire field of view of the user **102** (in VR use case). In another example, the display **204** includes a touchscreen display configured to receive a user input via a contact on the touchscreen display.

The processor **206** includes an AR/VR application **214** and a 3DOF tracking system **108**. The AR/VR application **214** retrieves virtual content (e.g., 3D object model) based on a detection of the physical object **104** or physical environment. The AR/VR application **214** renders the virtual object in the display **204**. In one example embodiment, the AR/VR application **214** includes a local rendering engine that generates a visualization of virtual content overlaid (e.g., superimposed upon, or otherwise displayed in tandem with) on an image of the physical object **104** captured by the optical sensor **212**. A visualization of the virtual content may be manipulated by adjusting a position of the physical object **104** (e.g., its physical location, orientation, or both) relative to the AR/VR display device **106**. Similarly, the visualization of the virtual content may be manipulated by adjusting a pose of the AR/VR display device **106** relative to the physical object **104**. For a VR application, the AR/VR application **214** displays the virtual content in the display **204** at a location (in the display **204**) determined based on a pose of the AR/VR display device **106**.

The 3DOF tracking system **108** estimates a rotational pose of the AR/VR display device **106**. For example, the 3DOF tracking system **108** uses sensor signals from the inertial sensor **210** to track a location of the AR/VR display device **106** relative to a frame of reference (e.g., real world environment **110**). As such, the 3DOF tracking system **108** tracks whether the user **102** of the AR/VR display device **106** is looking left or right, rotating their head up or down, and pivoting/tilting left or right.

In one example embodiment, the 3DOF tracking system **108** is initialized under dynamic and non-stationary conditions. In other words, the 3DOF tracking system **108** instantly initializes and starts a 3DoF (gravity aligned) tracking experience under dynamic and non-stationary conditions. In one example, the present method relies on the assumption that people stay roughly on the same spot is valid for most use cases (sitting, walking). Example components of the 3DOF tracking system **108** are described in more detail below with respect to FIG. **3**.

The storage device **208** stores virtual content **216**. The virtual content **216** includes, for example, a database of visual references (e.g., images of physical objects) and corresponding experiences (e.g., three-dimensional virtual object models).

Any one or more of the modules described herein may be implemented using hardware (e.g., a processor of a machine) or a combination of hardware and software. For example, any module described herein may configure a processor to perform the operations described herein for that module. Moreover, any two or more of these modules may be combined into a single module, and the functions described herein for a single module may be subdivided among multiple modules. Furthermore, according to various

example embodiments, modules described herein as being implemented within a single machine, database, or device may be distributed across multiple machines, databases, or devices.

FIG. 3 is a block diagram illustrating the 3DOF tracking system **108** in accordance with one example embodiment. The 3DOF tracking system **108** includes a gyroscope module **302**, an accelerometer module **304**, an inclination and position module **306**.

The gyroscope module **302** retrieves gyroscope signal from the gyroscope **218**. The accelerometer module **304** retrieves accelerometer signal from the accelerometer **220**.

The inclination and position module **306** performs the dynamic initialization by accessing the accelerometer and gyroscope signals from the accelerometer module **304** and the gyroscope module **302** respectively. The inclination and position module **306** integrates orientation and position signals, constrains the position signal to remain 'close' to the (zero) starting position.

In one example embodiment, the inclination and position module **306** includes an instant initialization module **308**, an integration module **314**, and a continuous update module **310**. The instant initialization module **308** determines an initial state that indicates a starting position. The initial state comprises a combination of an initial orientation, an initial position, and an initial velocity of the 3DOF tracking system **108**.

The integration module **314** integrates the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state. In one example, the integration module **314** integrates the gyroscope signal to determine the orientation signal starting from the initial orientation, rotates an additional accelerometer signal from the accelerometer using the orientation signal, subtract gravity, and performs a double integration starting from the initial position to determine the position signal.

The continuous update module **310** refines an inclination signal of the orientation signal using the position signal. In one example, the continuous update module **310** corrects the gyroscope signal and the accelerometer signal with biases and refines the inclination signal and biases using the position signal. In another example, the continuous update module **310** accesses a magnetometer signal from the magnetometer **222** of the inertial sensor **210**. The continuous update module **310** refines the orientation signal using the magnetometer signal. In yet another example, the continuous update module **310** accesses a location signal from a radio-based sensor **312** of the AR/VR display device **106**. The continuous update module **310** refines the position signal using the location signal.

FIG. 4 is a flow diagram illustrating a method for adapting the inclination in accordance with one example embodiment. Operations in the method **400** may be performed by the 3DOF tracking system **108**, using components (e.g., modules, engines) described above with respect to FIG. 3. Accordingly, the method **400** is described by way of example with reference to the inclination and position module **306**. However, it shall be appreciated that at least some of the operations of the method **400** may be deployed on various other hardware configurations or be performed by similar components residing elsewhere.

In block **402**, the inclination and position module **306** determines an initial state (initial inclination, position, velocity). In one example, the initial state comprises an initial position and an initial velocity. In block **404**, the inclination and position module **306** integrates gyroscope and accelerometer signal to obtain orientation and position

at the next time instant using the initial state. In block **406**, the inclination and position module **306** adapts the inclination or state by constraining the position to remain close to the starting position.

It is to be noted that other embodiments may use different sequencing, additional or fewer operations, and different nomenclature or terminology to accomplish similar functions. In some embodiments, various operations may be performed in parallel with other operations, either in a synchronous or asynchronous manner. The operations described herein were chosen to illustrate some principles of operations in a simplified form.

FIG. 5 is a flow diagram illustrating a method for refining an inclination signal in accordance with one example embodiment. Operations in the method **500** may be performed by the 3DOF tracking system **108**, using components (e.g., modules, engines) described above with respect to FIG. 3. Accordingly, the method **500** is described by way of example with reference to the inclination and position module **306**. However, it shall be appreciated that at least some of the operations of the method **500** may be deployed on various other hardware configurations or be performed by similar components residing elsewhere.

In block **502**, the AR/VR display device **106** starts the 3DOF tracking system **108**. In block **504**, the gyroscope module **302** accesses gyroscope signal from the gyroscope **218**. In block **506**, the accelerometer module **304** accesses acceleration signal from the accelerometer **220**. In block **508**, the instant initialization module **308** determines initial inclination without stationary initialization requirement using acceleration signal. In block **510**, the instant initialization module **308** accesses additional gyroscope signal and additional acceleration signal. In block **512**, the instant initialization module **308** integrates additional gyroscope signal to determine orientation signal starting from initial inclination and zero heading. In block **514**, the instant initialization module **308** rotates additional accelerometer signal using orientation signal, subtracts gravity, and performs double integration to determine position signal (starting from zero). In block **516**, the continuous update module **310** refines inclination based on the position signal.

FIG. 6 illustrates a routine **600** in accordance with one embodiment. In block **602**, routine **600** accesses a gyroscope signal from a gyroscope of the 3DOF tracking device. In block **604**, routine **600** accesses an accelerometer signal from an accelerometer of the 3DOF tracking device. In block **606**, routine **600** determines an initial state indicating an initial orientation of the 3DOF tracking device. In block **608**, routine **600** integrates the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state, the orientation and position signals indicating a starting position. In block **610**, routine **600** refines an inclination signal of the orientation signal using the position signal.

FIG. 7 is a block diagram illustrating an example algorithm for initialization in accordance with one example embodiment.

System with Head-Wearable Apparatus

FIG. 8 illustrates a network environment **800** in which the head-wearable apparatus **802** can be implemented according to one example embodiment. FIG. 8 is a high-level functional block diagram of an example head-wearable apparatus **802** communicatively coupled a mobile client device **838** and a server system **832** via various network **840**.

head-wearable apparatus **802** includes a camera, such as at least one of visible light camera **812**, infrared emitter **814** and infrared camera **816**. The client device **838** can be capable of connecting with head-wearable apparatus **802** using both a communication **834** and a communication **836**. client device **838** is connected to server system **832** and network **840**. The network **840** may include any combination of wired and wireless connections.

The head-wearable apparatus **802** further includes two image displays of the image display of optical assembly **804**. The two include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus **802**. The head-wearable apparatus **802** also includes image display driver **808**, image processor **810**, low-power low power circuitry **826**, and high-speed circuitry **818**. The image display of optical assembly **804** are for presenting images and videos, including an image that can include a graphical user interface to a user of the head-wearable apparatus **802**.

The image display driver **808** commands and controls the image display of the image display of optical assembly **804**. The image display driver **808** may deliver image data directly to the image display of the image display of optical assembly **804** for presentation or may have to convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H. 264 (MPEG-4 Part 10), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to compression formats such as Portable Network Group (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (Exif) or the like.

As noted above, head-wearable apparatus **802** includes a frame and stems (or temples) extending from a lateral side of the frame. The head-wearable apparatus **802** further includes a user input device **806** (e.g., touch sensor or push button) including an input surface on the head-wearable apparatus **802**. The user input device **806** (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

The components shown in FIG. 8 for the head-wearable apparatus **802** are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively, or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus **802**. Left and right can include digital camera elements such as a complementary metal-oxide-semiconductor (CMOS) image sensor, charge coupled device, a camera lens, or any other respective visible or light capturing elements that may be used to capture data, including images of scenes with unknown objects.

The head-wearable apparatus **802** includes a memory **822** which stores instructions to perform a subset or all of the functions described herein. memory **822** can also include storage device.

As shown in FIG. 8, high-speed circuitry **818** includes high-speed processor **820**, memory **822**, and high-speed wireless circuitry **824**. In the example, the image display driver **808** is coupled to the high-speed circuitry **818** and operated by the high-speed processor **820** in order to drive the left and right image displays of the image display of optical assembly **804**. high-speed processor **820** may be any processor capable of managing high-speed communications and operation of any general computing system needed for

head-wearable apparatus **802**. The high-speed processor **820** includes processing resources needed for managing high-speed data transfers on communication **836** to a wireless local area network (WLAN) using high-speed wireless circuitry **824**. In certain examples, the high-speed processor **820** executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus **802** and the operating system is stored in memory **822** for execution. In addition to any other responsibilities, the high-speed processor **820** executing a software architecture for the head-wearable apparatus **802** is used to manage data transfers with high-speed wireless circuitry **824**. In certain examples, high-speed wireless circuitry **824** is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as Wi-Fi. In other examples, other high-speed communications standards may be implemented by high-speed wireless circuitry **824**.

The low power wireless circuitry **830** and the high-speed wireless circuitry **824** of the head-wearable apparatus **802** can include short range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). The client device **838**, including the transceivers communicating via the communication **834** and communication **836**, may be implemented using details of the architecture of the head-wearable apparatus **802**, as can other elements of network **840**.

The memory **822** includes any storage device capable of storing various data and applications, including, among other things, camera data generated by the left and right, infrared camera **816**, and the image processor **810**, as well as images generated for display by the image display driver **808** on the image displays of the image display of optical assembly **804**. While memory **822** is shown as integrated with high-speed circuitry **818**, in other examples, memory **822** may be an independent standalone element of the head-wearable apparatus **802**. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor **820** from the image processor **810** or low power processor **828** to the memory **822**. In other examples, the high-speed processor **820** may manage addressing of memory **822** such that the low power processor **828** will boot the high-speed processor **820** any time that a read or write operation involving memory **822** is needed.

As shown in FIG. 8, the low power processor **828** or high-speed processor **820** of the head-wearable apparatus **802** can be coupled to the camera (visible light camera **812**; infrared emitter **814**, or infrared camera **816**), the image display driver **808**, the user input device **806** (e.g., touch sensor or push button), and the memory **822**.

The head-wearable apparatus **802** is connected with a host computer. For example, the head-wearable apparatus **802** is paired with the client device **838** via the communication **836** or connected to the server system **832** via the network **840**. server system **832** may be one or more computing devices as part of a service or network computing system, for example, that include a processor, a memory, and network communication interface to communicate over the network **840** with the client device **838** and head-wearable apparatus **802**.

The client device **838** includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network **840**, communication **834** or communication **836**. client device **838** can further store at least portions

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of the instructions for generating a binaural audio content in the client device **838**'s memory to implement the functionality described herein.

Output components of the head-wearable apparatus **802** include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver **808**. The output components of the head-wearable apparatus **802** further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus **802**, the client device **838**, and server system **832**, such as the user input device **806**, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

The head-wearable apparatus **802** may optionally include additional peripheral device elements. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with head-wearable apparatus **802**. For example, peripheral device elements may include any I/O components including output components, motion components, position components, or any other such elements described herein.

For example, the biometric components include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), WiFi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over and communication **836** from the client device **838** via the low power wireless circuitry **830** or high-speed wireless circuitry **824**.

Where a phrase similar to "at least one of A, B, or C," "at least one of A, B, and C," "one or more A, B, or C," or "one or more of A, B, and C" is used, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C.

Changes and modifications may be made to the disclosed embodiments without departing from the scope of the present disclosure. These and other changes or modifications are

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intended to be included within the scope of the present disclosure, as expressed in the following claims.

FIG. **9** is a block diagram **900** illustrating a software architecture **904**, which can be installed on any one or more of the devices described herein. The software architecture **904** is supported by hardware such as a machine **902** that includes Processors **920**, memory **926**, and I/O Components **938**. In this example, the software architecture **904** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **904** includes layers such as an operating system **912**, libraries **910**, frameworks **908**, and applications **906**. Operationally, the applications **906** invoke API calls **950** through the software stack and receive messages **952** in response to the API calls **950**.

The operating system **912** manages hardware resources and provides common services. The operating system **912** includes, for example, a kernel **914**, services **916**, and drivers **922**. The kernel **914** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **914** provides memory management, Processor management (e.g., scheduling), Component management, networking, and security settings, among other functionality. The services **916** can provide other common services for the other software layers. The drivers **922** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **922** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

The libraries **910** provide a low-level common infrastructure used by the applications **906**. The libraries **910** can include system libraries **918** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **910** can include API libraries **924** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **910** can also include a wide variety of other libraries **928** to provide many other APIs to the applications **906**.

The frameworks **908** provide a high-level common infrastructure that is used by the applications **906**. For example, the frameworks **908** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **908** can provide a broad spectrum of other APIs that can be used by the applications **906**, some of which may be specific to a particular operating system or platform.

In an example embodiment, the applications **906** may include a home application **936**, a contacts application **930**, a browser application **932**, a book reader application **934**, a location application **942**, a media application **944**, a messaging application **946**, a game application **948**, and a broad assortment of other applications such as a third-party appli-

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cation **940**. The applications **906** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **906**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **940** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **940** can invoke the API calls **950** provided by the operating system **912** to facilitate functionality described herein.

FIG. **10** is a diagrammatic representation of the machine **1000** within which instructions **1008** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1008** may cause the machine **1000** to execute any one or more of the methods described herein. The instructions **1008** transform the general, non-programmed machine **1000** into a particular machine **1000** programmed to carry out the described and illustrated functions in the manner described. The machine **1000** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a PDA, an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1008**, sequentially or otherwise, that specify actions to be taken by the machine **1000**. Further, while only a single machine **1000** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1008** to perform any one or more of the methodologies discussed herein.

The machine **1000** may include Processors **1002**, memory **1004**, and I/O Components **1042**, which may be configured to communicate with each other via a bus **1044**. In an example embodiment, the Processors **1002** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an ASIC, a Radio-Frequency Integrated Circuit (RFIC), another Processor, or any suitable combination thereof) may include, for example, a Processor **1006** and a Processor **1010** that execute the instructions **1008**. The term “Processor” is intended to include multi-core Processors that may comprise two or more independent Processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **10** shows multiple Processors **1002**, the machine **1000** may include a single Processor with a single core, a single Processor with multiple cores (e.g., a multi-

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core Processor), multiple Processors with a single core, multiple Processors with multiples cores, or any combination thereof.

The memory **1004** includes a main memory **1012**, a static memory **1014**, and a storage unit **1016**, both accessible to the Processors **1002** via the bus **1044**. The main memory **1004**, the static memory **1014**, and storage unit **1016** store the instructions **1008** embodying any one or more of the methodologies or functions described herein. The instructions **1008** may also reside, completely or partially, within the main memory **1012**, within the static memory **1014**, within machine-readable medium **1018** within the storage unit **1016**, within at least one of the Processors **1002** (e.g., within the Processor’s cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**.

The I/O Components **1042** may include a wide variety of Components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O Components **1042** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O Components **1042** may include many other Components that are not shown in FIG. **10**. In various example embodiments, the I/O Components **1042** may include output Components **1028** and input Components **1030**. The output Components **1028** may include visual Components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic Components (e.g., speakers), haptic Components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input Components **1030** may include alphanumeric input Components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input Components), point-based input Components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input Components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input Components), audio input Components (e.g., a microphone), and the like.

In further example embodiments, the I/O Components **1042** may include biometric Components **1032**, motion Components **1034**, environmental Components **1036**, or position Components **1038**, among a wide array of other Components. For example, the biometric Components **1032** include Components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion Components **1034** include acceleration sensor Components (e.g., accelerometer), gravitation sensor Components, rotation sensor Components (e.g., gyroscope), and so forth. The environmental Components **1036** include, for example, illumination sensor Components (e.g., photometer), temperature sensor Components (e.g., one or more thermometers that detect ambient temperature), humidity sensor Components, pressure sensor Components (e.g., barometer), acoustic sensor Components (e.g., one or more microphones that detect background noise), proximity sen-

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sor Components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other Components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position Components **1038** include location sensor Components (e.g., a GPS receiver Component), altitude sensor Components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor Components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O Components **1042** further include communication Components **1040** operable to couple the machine **1000** to a network **1020** or devices **1022** via a coupling **1024** and a coupling **1026**, respectively. For example, the communication Components **1040** may include a network interface Component or another suitable device to interface with the network **1020**. In further examples, the communication Components **1040** may include wired communication Components, wireless communication Components, cellular communication Components, Near Field Communication (NFC) Components, Bluetooth® Components (e.g., Bluetooth® Low Energy), Wi-Fi® Components, and other communication Components to provide communication via other modalities. The devices **1022** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication Components **1040** may detect identifiers or include Components operable to detect identifiers. For example, the communication Components **1040** may include Radio Frequency Identification (RFID) tag reader Components, NFC smart tag detection Components, optical reader Components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Data-glyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection Components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication Components **1040**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., memory **1004**, main memory **1012**, static memory **1014**, and/or memory of the Processors **1002**) and/or storage unit **1016** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1008**), when executed by Processors **1002**, cause various operations to implement the disclosed embodiments.

The instructions **1008** may be transmitted or received over the network **1020**, using a transmission medium, via a network interface device (e.g., a network interface Component included in the communication Components **1040**) and using any one of a number of well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1008** may be transmitted or received using a transmission medium via the coupling **1026** (e.g., a peer-to-peer coupling) to the devices **1022**.

As used herein, the terms “Machine-Storage Medium,” “device-storage medium,” and “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms refer to a single or multiple

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storage devices and/or media (e.g., a centralized or distributed database, and/or associated caches and servers) that store executable instructions and/or data. The terms shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of Machine-Storage Media, computer-storage media, and/or device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), field-programmable gate arrays (FPGAs), and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “Machine-Storage Media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium” discussed below.

The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure. The terms “transmission medium” and “signal medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying the instructions **1416** for execution by the machine **1400**, and include digital or analog communications signals or other intangible media to facilitate communication of such software. Hence, the terms “transmission medium” and “signal medium” shall be taken to include any form of modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

The terms “machine-readable medium,” “Computer-Readable Medium,” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure. The terms are defined to include both Machine-Storage Media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals.

Although an embodiment has been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof, show by way of illustration, and not of limitation, specific embodiments in which the subject matter may be practiced. The embodiments illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

Such embodiments of the inventive subject matter may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept if more than one is in fact disclosed. Thus, although specific embodiments have been illustrated and described herein, it should be

appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

EXAMPLES

Example 1 is a method for dynamically initializing a 3 degrees of freedom (3DOF) tracking device comprising: accessing a gyroscope signal from a gyroscope of the 3DOF tracking device; accessing an accelerometer signal from an accelerometer of the 3DOF tracking device; determining an initial state comprising a combination of an initial orientation, an initial position, and an initial velocity of the 3DOF tracking device, the initial state indicating a starting condition of the 3DOF tracking device; integrating the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state; and refining an inclination signal of the orientation signal using the position signal.

Example 2 includes example 1, wherein refining the inclination signal comprises: constraining the position signal to remain substantially close to the starting position.

Example 3 includes example 1, wherein determining the initial state comprises: determining the initial state is based on the accelerometer signal without requiring the 3DOF tracking device to remain in stationary conditions.

Example 4 includes example 1, wherein determining the initial state comprises: using previously estimated orientation and positional signals of the 3DOF tracking device.

Example 5 includes example 1, wherein integrating the gyroscope signal and the accelerometer signal comprise: integrating the gyroscope signal to determine the orientation signal starting from the initial orientation; rotating the accelerometer signal from the accelerometer using the orientation signal; subtracting gravity; and performing a double integration starting from the initial position to determine the position signal.

Example 6 includes example 1, further comprising: correcting the gyroscope signal and the accelerometer signal with biases; and refining the inclination signal and biases using the position signal.

Example 7 includes example 1, further comprising: accessing a magnetometer signal from a magnetometer of the 3DOF tracking device; and refining the orientation signal using the magnetometer signal.

Example 8 includes example 1, further comprising: accessing a location signal from a radio-based sensor of the 3DOF tracking device; and refining the position signal using the location signal.

Example 9 includes example 1, further comprising: detecting that the 3DOF tracking device is switched on; and in response to detecting that the 3DOF tracking device is switched on, initializing the 3DOF tracking device without waiting for the stationary condition of the 3DOF tracking device.

Example 10 includes example 9, further comprising: continuously determine the inclination after detecting that the 3DOF tracking device is turned on.

Example 11 is a computing apparatus comprising: a processor; and a memory storing instructions that, when executed by the processor, configure the apparatus to perform operations comprising: accessing a gyroscope signal from a gyroscope of a 3DOF tracking device; accessing an accelerometer signal from an accelerometer of the 3DOF tracking device; determining an initial state comprising a combination of an initial orientation, an initial position, and an initial velocity of the 3DOF tracking device, the initial state indicating a starting condition of the 3DOF tracking device; integrating the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state; and refining an inclination signal of the orientation signal using the position signal.

Example 12 includes example 11, wherein refining the inclination signal comprises: constrain the position signal to remain substantially close to the starting position.

Example 13 includes example 11, wherein determining the initial state comprises: determine the initial state is based on the accelerometer signal without requiring the 3DOF tracking device to remain in stationary conditions.

Example 14 includes example 11, wherein determining the initial state comprises: use previously estimated orientation and positional signals of the 3DOF tracking device.

Example 15 includes example 11, wherein integrating the gyroscope signal and the accelerometer signal comprise: integrate the gyroscope signal to determine the orientation signal starting from the initial orientation; rotate the accelerometer signal from the accelerometer using the orientation signal; subtract gravity; and perform a double integration starting from the initial position to determine the position signal.

Example 16 includes example 11, wherein the instructions further configure the apparatus to: correct the gyroscope signal and the accelerometer signal with biases; and refine the inclination signal and biases using the position signal.

Example 17 includes example 11, wherein the instructions further configure the apparatus to: access a magnetometer signal from a magnetometer of the 3DOF tracking device; and refine the orientation signal using the magnetometer signal.

Example 18 includes example 11, wherein the instructions further configure the apparatus to: access a location signal from a radio-based sensor of the 3DOF tracking device; and refine the position signal using the location signal.

Example 19 includes example 11, wherein the instructions further configure the apparatus to: detect that the 3DOF tracking device is switched on; in response to detecting that the 3DOF tracking device is switched on, initializing the 3DOF tracking device without waiting for the stationary condition of the 3DOF tracking device; and continuously determine the inclination after detecting that the 3DOF tracking device is turned on.

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Example 20 is a non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising: accessing a gyroscope signal from a gyroscope of a 3DOF tracking device; accessing an accelerometer signal from an accelerometer of the 3DOF tracking device; determining an initial state comprising a combination of an initial orientation, an initial position, and an initial velocity of the 3DOF tracking device, the initial state indicating a starting condition of the 3DOF tracking device; integrating the gyroscope signal and the accelerometer signal to obtain orientation and position signals using the initial state; and refining an inclination signal of the orientation signal using the position signal.

What is claimed is:

1. A method comprising:
turning on a 3 degrees of freedom (3DOF) tracking device;
in response to turning on the 3DOF tracking device, performing a dynamic initialization process of the 3DOF tracking device by determining an initial state that indicates a dynamic and non-stationary condition of the 3DOF tracking device, the initial state corresponding with turning on the 3DOF tracking device; and
operating the 3DOF tracking device based on the dynamic and non-stationary condition of the 3DOF tracking device after the dynamic initialization process.
2. The method of claim 1, wherein the initial state comprises a combination of an initial orientation of the 3DOF tracking device, an initial position of the 3DOF tracking device, and an initial velocity of the 3DOF tracking device, wherein the 3DOF tracking device is part of an augmented reality device.
3. The method of claim 2, wherein determining the initial state further comprises:
in response to turning on the 3DOF tracking device, accessing a gyroscope signal from a gyroscope of the 3DOF tracking device;
in response to turning on the 3DOF tracking device, accessing an accelerometer signal from an accelerometer of the 3DOF tracking device;
integrating the gyroscope signal and the accelerometer signal to obtain an orientation signal and a position signal using the initial state;
refining an inclination signal of the orientation signal using the position signal; and
constraining the position signal to remain substantially close to the initial position of the 3DOF tracking device.
4. The method of claim 3, wherein determining the initial state comprises:
determining the initial state is based on the accelerometer signal without requesting a user of the 3DOF tracking device to remain in stationary conditions.
5. The method of claim 3, wherein determining the initial state comprises:
using previously estimated orientation and positional signals of the 3DOF tracking device.
6. The method of claim 3, wherein integrating the gyroscope signal and the accelerometer signal comprise:
integrating the gyroscope signal to determine the orientation signal starting from the initial orientation;
rotating the accelerometer signal from the accelerometer using the orientation signal;
subtracting gravity; and

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performing a double integration starting from the initial position to determine the position signal.

7. The method of claim 3, further comprising:
after integrating only the gyroscope signal and the accelerometer signal, correcting the gyroscope signal and the accelerometer signal with biases; and
refining the inclination signal and biases using the position signal.

8. The method of claim 3, further comprising:
after integrating only the gyroscope signal and the accelerometer signal, accessing a magnetometer signal from a magnetometer of the 3DOF tracking device; and
refining the orientation signal using the magnetometer signal.

9. The method of claim 3, further comprising:
after integrating only the gyroscope signal and the accelerometer signal, accessing a location signal from a radio-based sensor of the 3DOF tracking device; and
refining the position signal using the location signal.

10. The method of claim 1, further comprising:
detecting that the 3DOF tracking device is switched on; and

in response to detecting that the 3DOF tracking device is switched on, initializing the 3DOF tracking device without waiting for a stationary condition of the 3DOF tracking device.

11. A computing apparatus comprising:
a processor; and
a memory storing instructions that, when executed by the processor, configure the apparatus to perform operations comprising:
turning on a 3 degrees of freedom (3DOF) tracking device;

in response to turning on the 3DOF tracking device, performing a dynamic initialization process of the 3DOF tracking device by determining an initial state that indicates a dynamic and non-stationary condition of the 3DOF tracking device, the initial state corresponding with turning on the 3DOF tracking device; and

operating the 3DOF tracking device based on the dynamic and non-stationary condition of the 3DOF tracking device after the dynamic initialization process.

12. The computing apparatus of claim 11, wherein the initial state comprises a combination of an initial orientation of the 3DOF tracking device, an initial position of the 3DOF tracking device, and an initial velocity of the 3DOF tracking device, wherein the 3DOF tracking device is part of an augmented reality device.

13. The computing apparatus of claim 12, wherein determining the initial state further comprises:

accessing a gyroscope signal from a gyroscope of the 3DOF tracking device;
accessing an accelerometer signal from an accelerometer of the 3DOF tracking device;
integrating the gyroscope signal and the accelerometer signal to obtain an orientation signal and a position signal using the initial state;
refining an inclination signal of the orientation signal using the position signal; and
constraining the position signal to remain substantially close to the initial position of the 3DOF tracking device.

14. The computing apparatus of claim 13, wherein determining the initial state comprises:

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determining the initial state is based on the accelerometer signal without requesting a user of the 3DOF tracking device to remain in stationary conditions.

15. The computing apparatus of claim 13, wherein determining the initial state comprises:

using previously estimated orientation and positional signals of the 3DOF tracking device.

16. The computing apparatus of claim 13, wherein integrating the gyroscope signal and the accelerometer signal comprise:

integrating the gyroscope signal to determine the orientation signal starting from the initial orientation;

rotating the accelerometer signal from the accelerometer using the orientation signal;

subtracting gravity; and

performing a double integration starting from the initial position to determine the position signal.

17. The computing apparatus of claim 13, wherein the operations further comprise:

after integrating only the gyroscope signal and the accelerometer signal, correcting the gyroscope signal and the accelerometer signal with biases, and

refining the inclination signal and biases using the position signal.

18. The computing apparatus of claim 13, wherein the operations further comprise:

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after integrating only the gyroscope signal and the accelerometer signal, accessing a magnetometer signal from a magnetometer of the 3DOF tracking device; and refining the orientation signal using the magnetometer signal.

19. The computing apparatus of claim 13, wherein the operations further comprise:

after integrating only the gyroscope signal and the accelerometer signal, accessing a location signal from a radio-based sensor of the 3DOF tracking device; and refining the position signal using the location signal.

20. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising:

turning on a 3 degrees of freedom (3DOF) tracking device;

in response to turning on the 3DOF tracking device, performing a dynamic initialization process of the 3DOF tracking device by determining an initial state that indicates a dynamic and non-stationary condition of the 3DOF tracking device, the initial state corresponding with turning on the 3DOF tracking device; and

operating the 3DOF tracking device based on the dynamic and non-stationary condition of the 3DOF tracking device after the dynamic initialization process.

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